

HW7

2025-04-04

Problem 1 #Part a We want to learn about how alcohol consumption and smoking status predict cancer. So, the outcome here is cancer_tot, the number of individuals in the sample with the given alcohol and smoke status who have cancer. This is what a binomial distribution has, binomial outcomes is where we count the number of successes out of total. So a binomial model would be appropriate for this data.

$$\text{logit}(p) = \beta_0 + \beta_1 \text{alcohol} + \beta_2 \text{smoke}$$

#Part b

```
library(readxl)
cancer_data <- read_excel("/Users/manideependyala/Downloads/cancer_dat.xlsx")
print(cancer_data)
```

```
## # A tibble: 4 x 4
##   alcohol smoke cancer_tot tot
##   <dbl> <dbl>   <dbl> <dbl>
## 1      0      0       10    48
## 2      0      1       33   103
## 3      1      0       16    63
## 4      1      1      424   716
```

```
bimodel <- glm(cbind(cancer_tot, tot - cancer_tot) ~ alcohol + smoke,
               data = cancer_data, family = binomial)

summary(bimodel)
```

```
##
## Call:
## glm(formula = cbind(cancer_tot, tot - cancer_tot) ~ alcohol +
##      smoke, family = binomial, data = cancer_data)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -1.7959      0.2676  -6.712 1.92e-11 ***
## alcohol        0.9687      0.2010   4.819 1.45e-06 ***
## smoke         1.1819      0.2420   4.883 1.04e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 70.0274  on 3  degrees of freedom
## Residual deviance:  2.8027  on 1  degrees of freedom
```

```
## AIC: 29.001
##
## Number of Fisher Scoring iterations: 4
```

#Part c Hypothesis test at the 0.01 level for whether alcohol consumption and smoking can be removed from the model. Null hypothesis : $\beta_{alcohol} = 0; \beta_{smoke} = 0$ From the above summary, Null deviance = 70.0274 (3 dof), Residual deviance = 2.8027 (1 dof) Test statistic, $D = \text{Null deviance} - \text{Residual deviance} = 67.2274$ $df = 3 - 1 = 2$, Under null hypothesis, it follows chi-square distribution with 2 (df) degrees of freedom. p-value = $1 - \chi^2_{(2)}(67.2274) = 2.553513e-15$ (see below for code). $p < 0.01$, so we reject null hypothesis. So we can say that atleast one of alcohol, smoke significantly improves the model prediction.

```
p_value <- 1 - pchisq(67.2274, df = 2)
p_value
```

```
## [1] 2.553513e-15
```

#Part d hypothesis test for whether alcohol is predictive of cancer status with smoke included in the model at the 0.05 level. Null hypothesis : $\beta_{alcohol} = 0$

```
coef_alcohol <- summary(bimodel)$coefficients["alcohol", ]
cat("z =", coef_alcohol["z value"],
    "\np-value =", coef_alcohol["Pr(>|z|)"])
```

```
## z = 4.818654
## p-value = 1.445299e-06
```

From summary of the model, estimate of alcohol is 0.9687, standard error is 0.2010. z-value is 4.819, p value is 1.45e-06. this is less than 0.05, so we reject null hypothesis. Alcohol is predictive of cancer status even when smoke is included.

#Part e Construct and interpret (in context) a 99% confidence interval for β_1 . 99 % CI is $\beta_1 \pm 2.576 \times SE$. Standard error is 0.2010 (from summary and part d), β_1 estimate for alcohol is 0.9687.

$$CI = [0.9687 \pm 2.576 \times 0.2010] = [0.4509, 1.4865]$$

At the 99% confidence level and keeping smoke predictor fixed, alcohol has an increase in the log odds of cancer between 0.451 and 1.487, which is them having about 1.57 to 4.42 times the odds of developing cancer than non drinkers.

#Part f True, a test at 0.05 level is 95% CI, if this doesn't include zero we reject null hypothesis at 0.05 level. But, 0.01 level is wider so it is possible to have zero in the CI, can happen when the p-value is just below the level value.

#Part g False, When data is aggregated properly and the model is correctly specified, fitting a binomial logistic regression on aggregated data is equivalent to fitting a logistic regression on the individual binary outcomes. Therefore, the test results should be the same.

Problem 2

```
library(faraway)
head(dvisits)
```

```
## sex age agesq income levyplus freepoor freerepa illness actdays hscore
## 1 1 0.19 0.0361 0.55 1 0 0 1 4 1
```

```
## 2  1 0.19 0.0361  0.45      1      0      0      1      2      1
## 3  0 0.19 0.0361  0.90      0      0      0      3      0      0
## 4  0 0.19 0.0361  0.15      0      0      0      1      0      0
## 5  0 0.19 0.0361  0.45      0      0      0      2      5      1
## 6  1 0.19 0.0361  0.35      0      0      0      5      1      9
##   chcond1 chcond2 doctorco nondocco hospadmi hospdays medicine prescrib
## 1      0      0      1      0      0      0      0      1      1
## 2      0      0      1      0      0      0      0      2      1
## 3      0      0      1      0      1      4      2      2      1
## 4      0      0      1      0      0      0      0      0      0
## 5      1      0      1      0      0      0      0      3      1
## 6      1      0      1      0      0      0      0      1      1
##   nonpresc
## 1      0
## 2      1
## 3      1
## 4      0
## 5      2
## 6      0
```

#Part a

```
plot_data <- na.omit(data.frame(illness = dvisits$illness, doctorco = dvisits$doctorco))

plot(plot_data$illness, plot_data$doctorco,
      xlab = "Illness", ylab = "Doctor's Consultations",
      main = "Doctorco vs. Illness", pch = 16)

lines(loess.smooth(plot_data$illness, plot_data$doctorco), col = "blue", lwd = 2)
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 1
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 1
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 1
```

```
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1

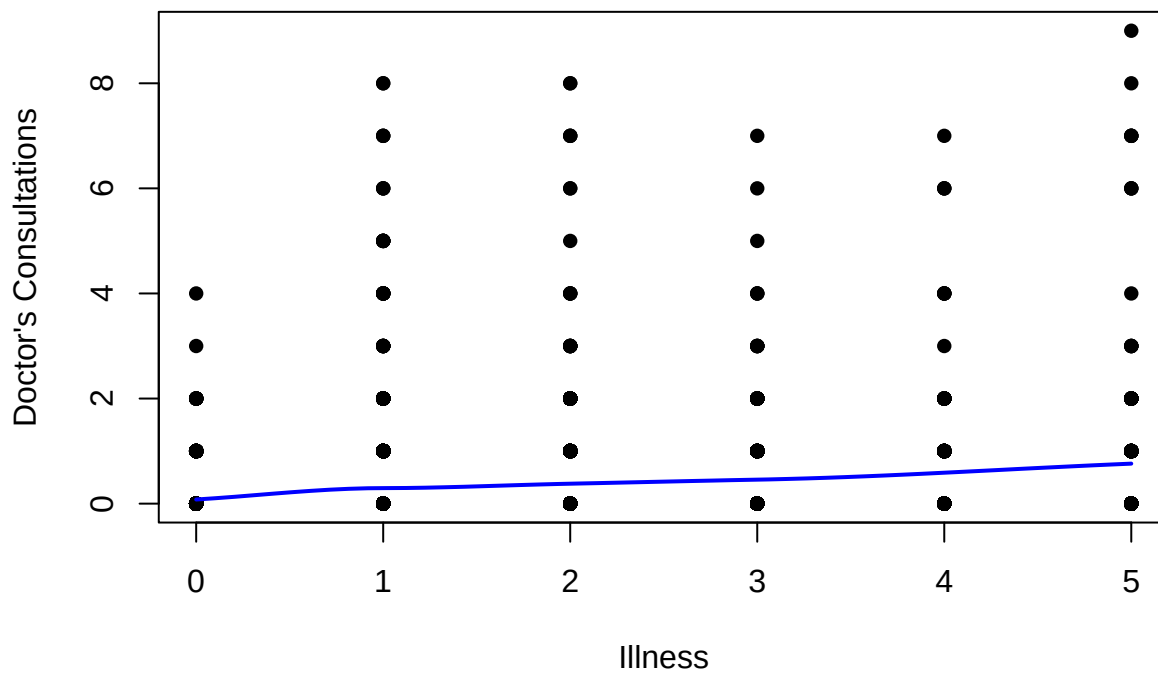
## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## pseudoinverse used at 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## neighborhood radius 1

## Warning in simpleLoess(y, x, w, span, degree = degree, parametric = FALSE, :
## reciprocal condition number 0
```

Doctorco vs. Illness



```

plot_data_age <- data.frame(age = dvisits$age, doctorco = dvisits$doctorco)
plot_data_age <- plot_data_age[is.finite(plot_data_age$age) & is.finite(plot_data_age$doctorco), ]

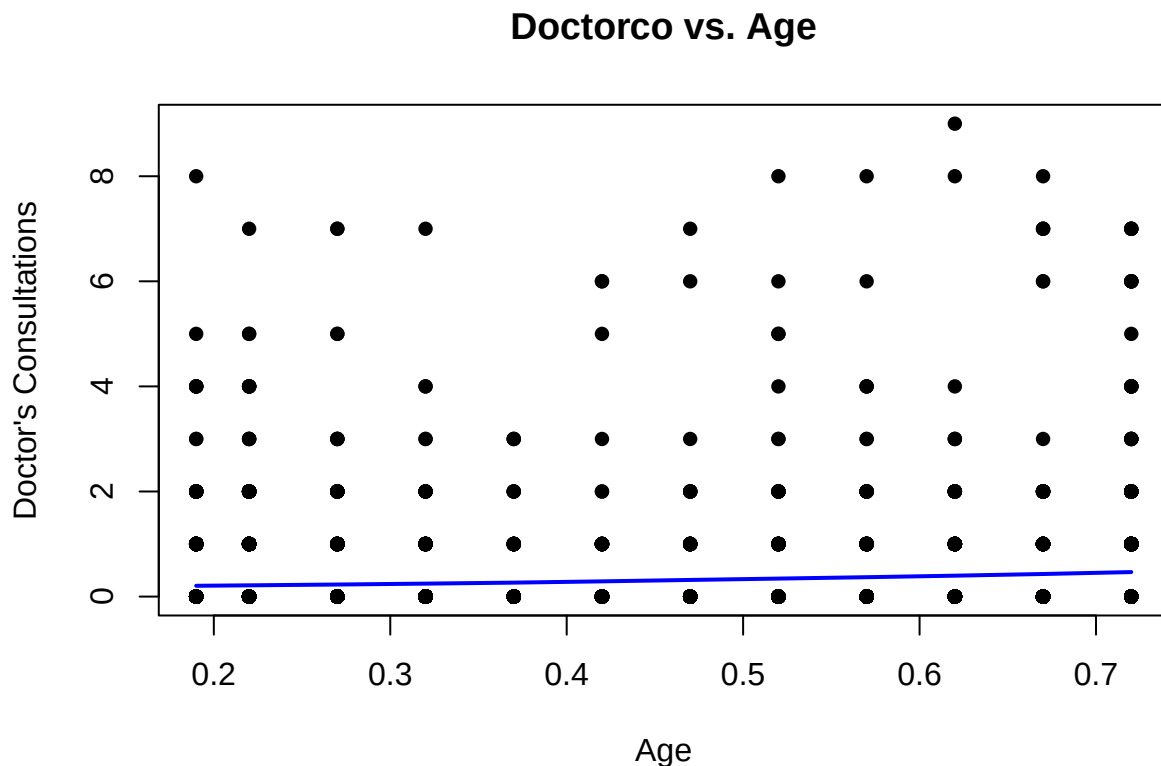
plot(plot_data_age$age, plot_data_age$doctorco,
     xlab = "Age", ylab = "Doctor's Consultations",
     main = "Doctorco vs. Age", pch = 16)

loess_model <- loess(doctorco ~ age, data = plot_data_age)

sorted_age <- sort(plot_data_age$age)
predicted_vals <- predict(loess_model, newdata = data.frame(age = sorted_age))

lines(sorted_age, predicted_vals, col = "blue", lwd = 2)

```



For the doctorco versus illness plot, the data show that as illness increases, the number of doctor consultations increase, there is variability at each illness level, so other factors may also influence the frequency of visits.

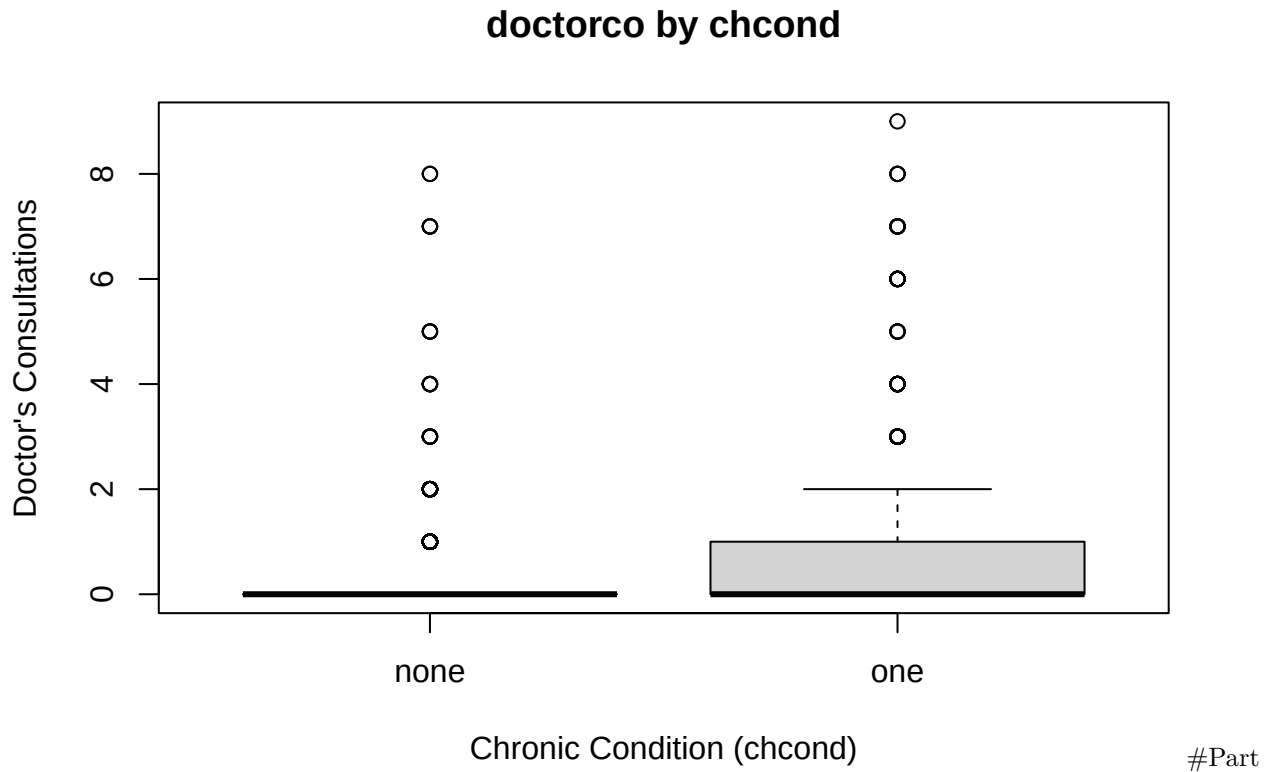
For the doctorco versus age plot, there is an upward trend indicating that older individuals generally have more doctor consultations, the relationship is not very steep and variation exists across the age range, so age alone does not fully explain the differences in consultation frequency. #Part b

```

dvisits$chcond <- with(dvisits,
                      factor(ifelse(chcond1 == 0 & chcond2 == 0, "none",
                                     ifelse(chcond1 + chcond2 == 1, "one", "both"))))

boxplot(doctorco ~ chcond, data = dvisits, # we chose boxplot because the covariate is categorical in t
       xlab = "Chronic Condition (chcond)",
       ylab = "Doctor's Consultations",
       main = "doctorco by chcond")

```



c

```
pomodel <- glm(doctorco ~ sex + age + agesq + income + illness + hscore + chcond,
               data = dvisits, family = poisson)
summary(pomodel)
```

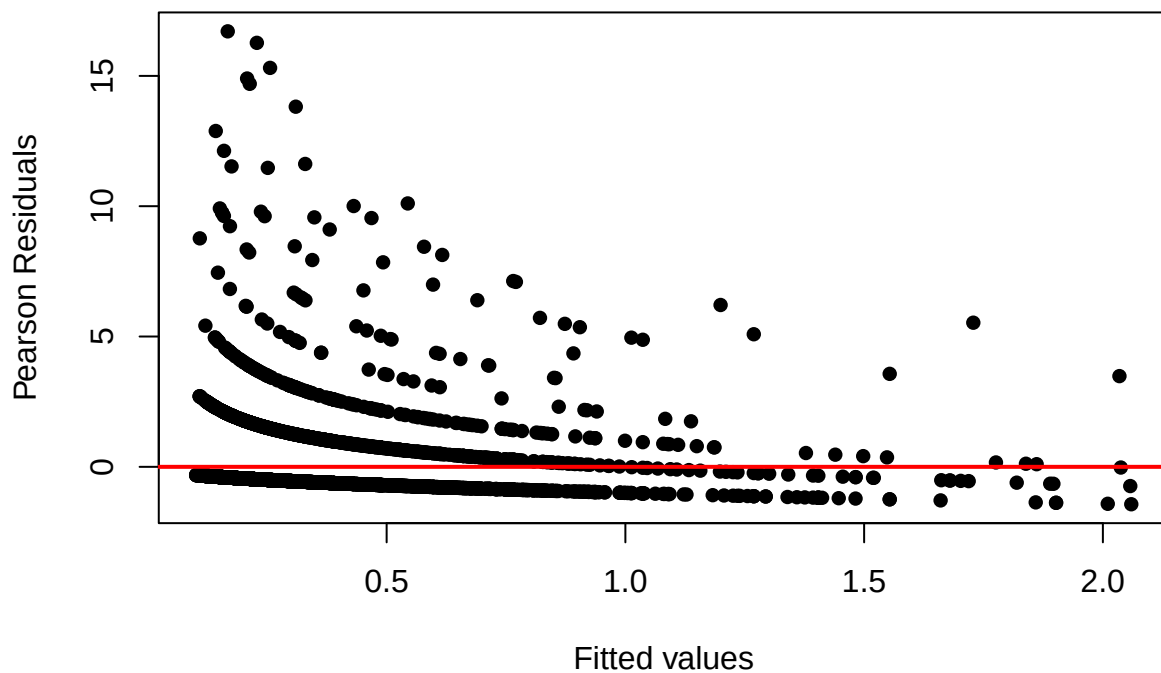
```
##
## Call:
## glm(formula = doctorco ~ sex + age + agesq + income + illness +
##      hscore + chcond, family = poisson, data = dvisits)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -2.397491   0.187112 -12.813  < 2e-16 ***
## sex          0.149728   0.055736   2.686  0.007223 **
## age          1.474072   0.992427   1.485  0.137459
## agesq       -0.749148   1.082574  -0.692  0.488933
## income      -0.121238   0.083633  -1.450  0.147160
## illness      0.225050   0.018309  12.292  < 2e-16 ***
## hscore       0.101185   0.009137  11.075  < 2e-16 ***
## chcondone    0.212097   0.063423   3.344  0.000825 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 5634.8  on 5189  degrees of freedom
## Residual deviance: 4955.6  on 5182  degrees of freedom
## AIC: 7303.1
##
```

```
## Number of Fisher Scoring iterations: 6
```

The income coefficient is -0.121238, meaning that one unit increase in income is associated with about a 12% decrease in the expected number of doctor visits ($\exp(-0.121) = 0.886$), keeping other factors fixed. This effect is not statistically significant ($p = 0.147$). #Part d

```
plot(pomodel$fitted.values, residuals(pomodel, type = "pearson"),
     xlab = "Fitted values", ylab = "Pearson Residuals",
     main = "Residual Plot for Poisson Regression", pch = 16)
abline(h = 0, col = "red", lwd = 2)
```

Residual Plot for Poisson Regression



Residuals appear larger at lower fitted values, indicating missing factors. They are not randomly scattered around zero, meaning the Poisson model may not fully capture the variation in the data.

#Part e

```
last_person <- dvisits[nrow(dvisits), ]
predicted_mean <- predict(pomodel, newdata = last_person, type = "response")
cat("Estimated mean number of doctor's consultations for the last person:",
    predicted_mean, "\n")
```

```
## Estimated mean number of doctor's consultations for the last person: 0.1729371
```

#Part f

```
lambda <- predicted_mean

prob_0 <- dpois(0, lambda)
prob_1 <- dpois(1, lambda)
```

```
prob_2 <- dpois(2, lambda)
```

```
cat("Probability of 0 consultations:", prob_0, "\n")
```

```
## Probability of 0 consultations: 0.8411905
```

```
cat("Probability of 1 consultation :", prob_1, "\n")
```

```
## Probability of 1 consultation : 0.145473
```

```
cat("Probability of 2 consultations:", prob_2, "\n")
```

```
## Probability of 2 consultations: 0.01257884
```

```
#Part g
```

```
pomodelr <- glm(doctorco ~ age + agesq + illness + hscore + chcond,  
               data = dvisits, family = poisson)
```

```
anova_result <- anova(pomodelr, pomodel, test = "Chisq")  
print(anova_result)
```

```
## Analysis of Deviance Table
```

```
##
```

```
## Model 1: doctorco ~ age + agesq + illness + hscore + chcond
```

```
## Model 2: doctorco ~ sex + age + agesq + income + illness + hscore + chcond
```

```
##   Resid. Df Resid. Dev Df Deviance Pr(>Chi)
```

```
## 1      5184      4966.9
```

```
## 2      5182      4955.6  2   11.375 0.003387 **
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Null hypothesis: $\beta_{sex} = \beta_{income} = 0$. $D = \text{Null} - \text{Residual deviance} = 4966.9 - 4955.6 = 11.375$, $df = 5184 - 5182 = 2$. chi-square gives a p-value of 0.003387 (< 0.1). So, we reject null hypothesis at 0.1 level. At least one of sex or income improves the model prediction if included in the model.

```
#Part h
```

```
coef_sex <- summary(pomodel)$coefficients["sex", ]
```

```
estimate_sex <- coef_sex["Estimate"]
```

```
se_sex <- coef_sex["Std. Error"]
```

```
z_critical <- qnorm(0.95)
```

```
CI_lower_sex <- estimate_sex - z_critical * se_sex
```

```
CI_upper_sex <- estimate_sex + z_critical * se_sex
```

```
cat("90% CI for the coefficient for sex: [", CI_lower_sex, ",", CI_upper_sex, "]\n")
```

```
## 90% CI for the coefficient for sex: [ 0.05805109 , 0.2414055 ]
```


#Part i Since the 90% confidence interval for the sex coefficient is $[0.05805109, 0.2414055]$ which does not include 0, we would reject null hypothesis at the 0.1 level, this shows that sex is a significant predictor. This aligns with the joint test where we got a deviance difference of 11.375 with 2 dof and a p-value of 0.003387, also leading us to reject null hypothesis at the 0.1 level. Together, these results indicate that including atleast one of sex, income improves the model's predictive ability.

Problem 3 #Part a The treatments are the two versions of the resume that differ only in applicant's name which shows the gender. One treatment is the resume with name : John (male), and the other is the identical resume with name : Jennifer (female).

#Part b

The experimental units are the nationwide sample of biology, chemistry, and physics professors who evaluated the resumes. There were 127 professors in total. The measurement units are the ratings provided by these professors in the student's competence and hireability, as well as the amount of salary and amount of mentoring they would offer the student.

#Part c

The responses are the evaluations given by the professors, which include: The perceived competence of the student, student's hireability, amount of salary they would offer, and amount of mentoring they would offer the student. These ratings given by the professors are the outcomes used to determine if gender bias influenced their judgments.

#Part d For each professor (experimental unit), there are two potential outcomes: If assigned a resume with male name, The professor would provide a set of ratings (for competence, hireability, salary, and mentoring) for the resume with the name John. If assigned a resume with female name, The same professor would provide a set of ratings for the resume named Jennifer.

Because each professor is only exposed to one version of the resume, The difference between outcomes shows the effect of the gender treatment on the evaluations.

#Part e The professors were randomly assigned to one of the two resumes. The description tells us that every participant received the same materials except for the student's name, which was randomly assigned with roughly equal groups (63 for the male and 64 for the female). They used simple randomization was used at the unit level of the individual professor.